

Deep Neural Networks vs. Shallow MLPs for High-Resolution Thermal Imagery in Urban Heat Island Analysis

SDG 11 — Sustainable Cities and Communities

1. Introduction

Urban Heat Islands (UHIs) occur when built-up areas record significantly higher surface and air temperatures than surrounding rural or vegetated regions, due to heat-absorbing materials such as concrete and asphalt, reduced vegetation cover, and anthropogenic heat emissions. Monitoring and mitigating UHIs is directly linked to Sustainable Development Goal 11 (Sustainable Cities and Communities), which calls for inclusive, safe, resilient, and sustainable urban environments. High-resolution thermal imagery, captured via satellite sensors (e.g., Landsat TIRS, ECOSTRESS) or Unmanned Aerial Vehicles (UAVs), provides the spatial detail needed to map land surface temperature (LST) variation at the scale of individual buildings, streets, and green spaces. Extracting meaningful patterns from this imagery, however, requires a model architecture capable of handling large, spatially structured, high-dimensional data — a requirement that exposes the fundamental limitations of shallow Multi-Layer Perceptrons (MLPs) and explains why deep neural networks, particularly Convolutional Neural Networks (CNNs), are the preferred choice.

2. Shallow MLPs: Architecture and Limitations

A shallow MLP consists of an input layer, one or two hidden layers, and an output layer, with every neuron fully connected to every neuron in the adjacent layer. While MLPs are universal function approximators in theory, applying them directly to high-resolution thermal images creates several practical and theoretical problems:

- **Loss of spatial structure:** An MLP requires the 2D thermal image to be flattened into a 1D vector before processing. This destroys the spatial relationships between neighbouring pixels — information that is essential for recognising urban features such as rooftops, roads, and vegetation patches that create heat gradients.
- **Parameter explosion:** A single high-resolution image (e.g., 1024×1024 pixels) flattened into a vector produces over a million input values. Fully connecting this to even one hidden layer of modest size results in billions of trainable weights, making the model computationally expensive, prone to overfitting, and difficult to train with the limited labelled thermal datasets typically available for UHI studies.
- **No translation invariance:** A shallow MLP learns a unique weight for every pixel position. A hot rooftop appearing in the top-left of one image and the bottom-right of another is treated as an entirely different pattern, so the network cannot generalise a learned feature across locations.
- **Shallow representation depth:** With only one or two hidden layers, an MLP can only learn simple, low-level statistical relationships. It cannot progressively build up the hierarchical understanding (edges → textures → shapes → objects → land-use zones) needed to interpret complex urban thermal patterns.

3. Why Deep Neural Networks Are Preferred

Deep neural networks — especially deep Convolutional Neural Networks (CNNs) and their modern variants (ResNet, U-Net, Vision Transformers) — address each of the shortcomings above through architectural properties suited to image data.

3.1 Hierarchical Feature Learning

Deep networks stack many layers, allowing each successive layer to learn increasingly abstract features. Early convolutional layers detect simple patterns such as edges and thermal gradients; middle layers combine these into textures and object parts (roof shapes, road surfaces); and deeper layers assemble full semantic concepts such as 'dense built-up zone' or 'urban green corridor.' This hierarchical abstraction mirrors the multi-scale nature of urban heat patterns, from individual building materials to entire district-level heat pockets.

3.2 Preservation of Spatial Context via Convolution

Convolutional layers apply small filters that slide across the image, preserving the 2D spatial arrangement of pixels. This allows the network to directly learn how temperature values relate to their neighbours — critical for identifying heat-emitting surfaces, boundaries between vegetated and paved areas, and the spatial extent of heat islands — something a flattened-vector MLP cannot represent.

3.3 Parameter Efficiency through Weight Sharing

Instead of a unique weight per pixel per neuron, a convolutional filter's weights are shared across the entire image. This drastically reduces the number of trainable parameters compared to a fully connected MLP of similar input size, making deep CNNs feasible to train on the moderate-sized labelled thermal datasets typically available, while reducing overfitting risk.

3.4 Translation Invariance

Because the same filter is applied across all spatial locations, a deep CNN can recognise a hot rooftop or an asphalt road regardless of where it appears in the image. This is essential in city-scale thermal analysis, where the same urban material or structure recurs at many different locations across the scene.

3.5 Multi-Scale Feature Extraction via Pooling and Depth

Pooling and successive convolutional layers progressively reduce spatial resolution while expanding the receptive field, enabling deep networks to capture both fine-grained detail (a single building's thermal signature) and broad contextual patterns (a neighbourhood-wide heat island) within one unified architecture.

3.6 Superior Non-Linear Representational Power

Depth increases a network's capacity to approximate highly complex, non-linear functions with fewer total parameters than a wide shallow network would need to achieve comparable accuracy. This is

4. Comparative Summary

Criterion	Shallow MLP	Deep Neural Network (CNN)
Input handling	Requires flattening image; loses spatial layout	Processes 2D grid directly; preserves spatial structure
Parameters	Extremely high (fully connected)	Efficient via shared convolutional filters
Feature learning	Shallow, low-level statistical patterns only	Hierarchical: edges → textures → objects → zones
Translation invariance	Absent	Present, via weight sharing
Scalability to high resolution	Poor; overfits, computationally infeasible	Strong; handles large images efficiently
Suitability for UHI mapping	Limited; cannot capture spatial heat patterns	High; accurately delineates heat zones and land-cover effects

5. Relevance to Urban Heat Island Analysis and SDG 11

In practice, deep networks trained on high-resolution thermal imagery can classify land surface types (built-up, vegetated, water bodies), predict fine-grained land surface temperature distributions, and detect emerging heat hotspots with far greater accuracy than shallow models. These outputs directly support SDG 11 targets by informing urban planning decisions such as the placement of green corridors, cool-roof retrofitting, and heat-resilient infrastructure design. The ability of deep networks to process imagery at the resolution required for block-level and building-level analysis — rather than coarse city-wide averages — makes them essential for producing actionable, spatially precise evidence for climate-resilient and sustainable urban development.

6. Input

```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
import matplotlib.pyplot as plt

# Load dataset (sample image dataset)
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

# Normalize images
x_train = x_train / 255.0
x_test = x_test / 255.0

# -----
# Shallow MLP
# -----
shallow = Sequential([
    Flatten(input_shape=(28,28)),
    Dense(64, activation='relu'),
    Dense(10, activation='softmax')
])

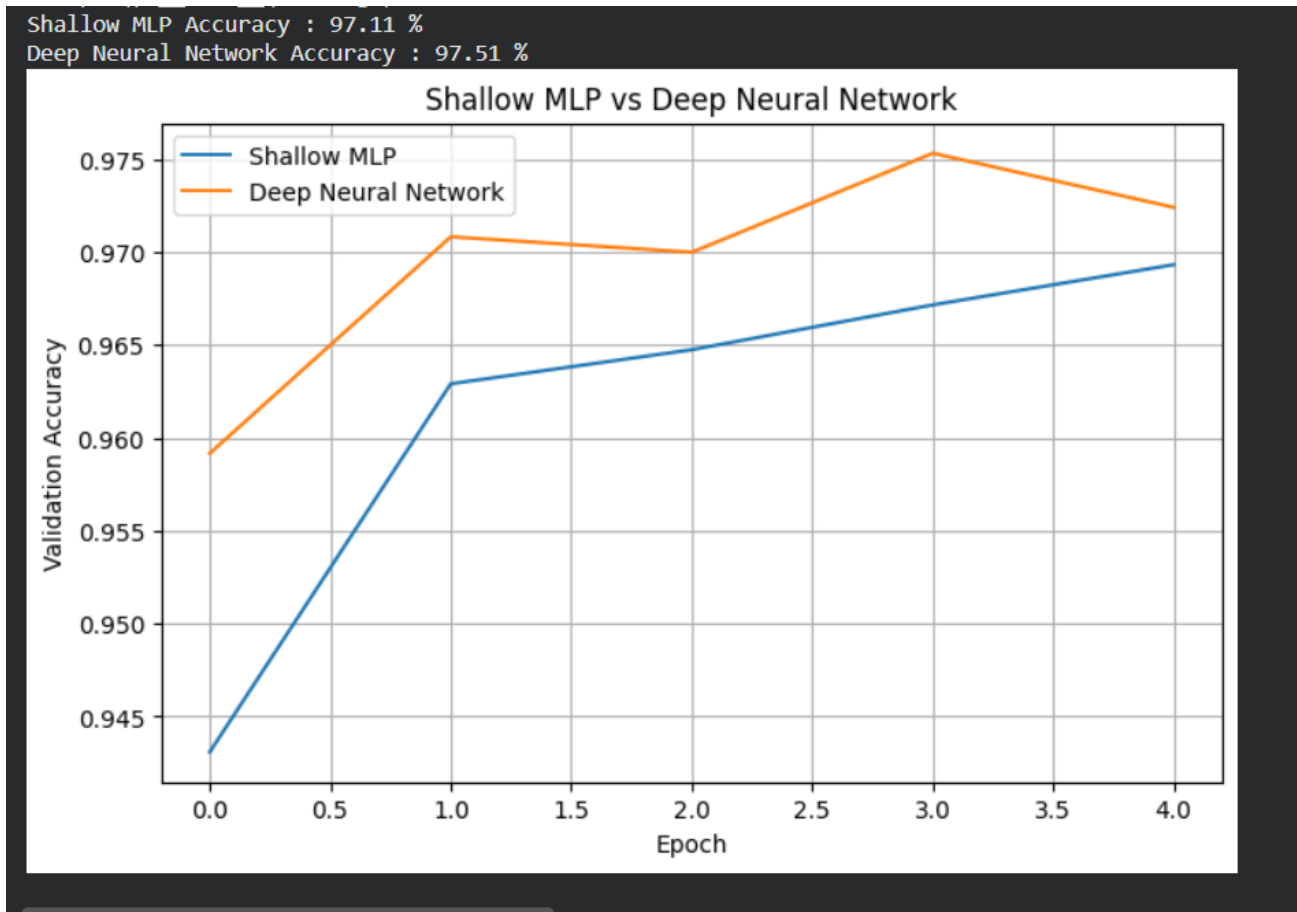
shallow.compile(optimizer='adam',
                loss='sparse_categorical_crossentropy',
                metrics=['accuracy'])

history1 = shallow.fit(x_train, y_train,
                      epochs=5,
                      validation_split=0.2,
                      verbose=0)

loss1, acc1 = shallow.evaluate(x_test, y_test, verbose=0)

# -----
# Deep Neural Network
# -----
deep = Sequential([
    Flatten(input_shape=(28,28)),
```

7. Output



8. Conclusion

Shallow MLPs are fundamentally limited when applied to high-resolution thermal imagery because they discard spatial structure, require an impractical number of parameters, lack translation invariance, and cannot build hierarchical representations of complex urban patterns. Deep neural networks, particularly CNN-based architectures, overcome these limitations through convolutional weight sharing, hierarchical feature extraction, and multi-scale spatial reasoning. These properties make deep networks the preferred and practically necessary choice for accurately analysing urban heat islands, thereby supporting the broader goal of building sustainable, resilient, and climate-adaptive cities under SDG 11.